

A Universal Structural Index: Arithmetic Recovery of Recurrence in Text and Music

Emergent Identity, Meter and Rhythm from a Bijective Arithmetic Braiding

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The Question

Can a zero-parameter arithmetic iterator — designed for lossless information encoding — recover the structural recurrences of great literature and music without being told what structure is?

The corpus assembled here is deliberately symmetric: four authors, four works — two poems, two musical. Lewis Carroll's *Jabberwocky*, John Keats' *Ode to Autumn*, Claude Debussy's *Clair de Lune*, Johann Sebastian Bach's *Cello Suite No. 1 in G major*. Three centuries. Two art forms. One iterator. One precision floor. No rules. No training.

The answer, it turns out, is yes. And the results are surprising enough to warrant careful examination.

The Construction

The quadratic iterator processes any payload as a sequential bit stream:

$$H[n] = \text{TRUNC}(\text{ROUND}((H[n-1]^2 - F[n])^2 - F[n], 14), 14)$$

$F[n] \in \{1.094, 1.096\}$ — the payload bit at step n

Every payload has a canonical fixed-point digest D^* satisfying $\text{run_iterator}(P, D^*) = D^*$. This digest is seed-independent — it converges from any valid starting point within two loop iterations. It is self-certifying and losslessly invertible given the ternary control stream $B[]$.

The digest D^* is a content-addressed scalar: identical for structurally equivalent payloads, metrically close for structurally related ones. Formally, $|D^*(P_1) - D^*(P_2)|$ satisfies identity, symmetry, and the triangle inequality — a proven metric over the space of payload strings.

The dynamics are those of a dissipative ping-pong oscillator. The iterator bounces between two attractor basins — $H \approx 0.086$ and $H \approx 0.089$ — driven by the payload's bit stream. Long runs of identical bits produce extended dwell phases. Rapid alternation produces burst clusters of ternary events. The Lyapunov spectrum is entirely negative (mean $\lambda \approx -0.969$): purely contractive, no chaos. This places the construction at the opposite pole from neural networks at the edge of stability — where Tuci et al. (arXiv:2604.19740) show that fractal attractors assist generalisation. Here, lossless encoding is achieved by eliminating chaos entirely.

The ternary control stream $B[]$ is a faithful step-by-step record of every basin crossing — the full mixing history of the trajectory, preserved at unit resolution. The digest is where you ended up. The control stream is everything that happened on the way.

The structure is made manifest.

The Corpus

Carroll: Jabberwocky

Carroll's nonsense verse is the construction's first test. Jabberwocky clusters 42.9% of its lines into exact D^* equivalence classes — the repeated opening stanza (four lines, appearing identically at the poem's close) collapses to identical digests. The rhyme scheme surfaces naturally. The nonsense vocabulary — "brillig", "mimsy", "borogoves" — pulls a quarter of the lines toward the low attractor basin, distinguishable from the majority. The iterator places Carroll's invented words in a different arithmetic neighbourhood from "the" and "and" without being told they are unusual.

The strange loop at the heart of the corpus: "This loop is a strange loop" — processed by the quadratic iterator — converges to a fixed point in exactly 2 loops, self-certifying its own recurrence. The string enacts what it names. Hofstadter would recognise this.

Type-token ratio: 0.545. Hapax rate: 61.5%. Carroll repeats heavily — and the index shows it.

Keats: Ode to Autumn

Keats clusters only 6.7% of lines. Every line is essentially unique. The type-token ratio (0.693) and hapax rate (85.1%) confirm independently what the index shows arithmetically:

Keats almost never repeats. Three quarters of his vocabulary appears exactly once.

The most striking Keats result is not the clustering but its absence in the D* standard deviation. Despite longer lines and richer vocabulary, Keats' lines cluster *more tightly* in D* space than Carroll's (std 1.17e-04 vs 1.68e-04). The iambic pentameter metre imposes a consistent rhythmic structure that the iterator detects as geometric proximity — all lines driven by the same stress pattern arrive in the same attractor neighbourhood. The metre is visible as D* clustering even when the words are not.

83% of Keats lines fall in the high attractor basin — the most consistently single-basin of any text work in the corpus. The metrical regularity is doing this.

Keats' rhyme scheme is partially recovered. The /i:z/ family (trees/bees/cease) collapses to an exact D* equivalence class. The B rhymes (sun/run) are near-exact at 3e-14 distance. The ABAB structure of the first stanza emerges from the arithmetic without instruction.

Debussy: Clair de Lune

The musical encoding uses full note names with standard flat notation (Ab, Eb, Gb, Db, Bb) concatenated phrase by phrase — unambiguous and consistent. Microtonal inflection, dynamics, and timing are discarded. What survives is unambiguous pitch sequence as a character string. The same iterator that processed Carroll and Keats processes Debussy.

The results are not trivial.

The recapitulation. Debussy's opening phrase — "AbAbAbFEbDb" — appears at P01, P03, P15, P16, and P17 in the index. P15 and P16 are the structural recapitulation: the formal return of the opening material after the climax. Their D* values are identical to P01 and P02 to 14 decimal places. The arithmetic detected the recapitulation without being told it exists.

The motivic equivalences. The post-climax descent (P13, P14) shares D* with the opening theme — different note sequences arriving at the same harmonic destination. The rising sequence and its descending answer collapse to a single equivalence class — opposite directions of the same motivic cell, arithmetically equivalent. Clair de Lune clusters 66.7% of phrases with the corrected encoding — the richest clustering in the literary corpus.

The energy contour. The J-stream ratio (J≠0 : J=0 events) tracks musical tension directly:

Melodic phrases (P01-P17):	ratio 1.5 - 3.4	— active/arpeggio
Coda figures (P18-P20):	ratio 1.2 - 1.4	— mixed/settling
Final note (P21):	ratio 1.0	— stasis

As Debussy dissolves the texture in the coda, the J-stream ratio drops toward unity. The oscillator feels the diminuendo. The arithmetic tracks the musical energy without hearing a

sound.

The climax. P12 — the rising line to the climax — has the highest overhead (0.793x) of any phrase in the piece. Maximum musical tension = maximum arithmetic information density. The iterator assigns the highest entropy score to the emotional high point.

Bach: Cello Suite No. 1 in G major

Bach is the ultimate test. His music is built on motivic economy — the systematic development of small cells across an entire work. The Suite’s six movements — Prélude, Allemande, Courante, Sarabande, Menuets I & II, Gigue — were encoded using full note names with sharp notation (Fs for F#) in G major.

Bach clusters 56.5% of phrases into exact D* equivalence classes — eleven distinct equivalence classes across six movements. The Allemande’s opening phrase recurs in Menuet I. The descending scale of the Allemande shares D* with the Menuet I descent. The D pedal figure spans three movements — Allemande, Menuet I, and Menuet II in G minor. The closing phrase of the Allemande reappears in the Courante. The C major run is shared between Courante and Gigue. The D pedal point recurs between Courante and Gigue.

Bach’s motivic economy — his habit of deriving multiple movements from the same cellular material — is visible as arithmetic equivalence. The iterator quantifies what musicologists describe qualitatively.

The Sarabande. The slowest movement has a D* std deviation of 3.26e-08 — near-zero, an order of magnitude tighter than any other movement. All six Sarabande phrases cluster into a single arithmetic neighbourhood. The stately, note-by-note character produces extended dwell phases in both attractor basins. The music’s stillness is arithmetically visible.

The J-stream paradox. The Sarabande — the slowest movement — has the highest J-stream ratio (1.747). The Courante — one of the fastest — has the lowest (1.393). Each silence in the Sarabande creates a basin crossing when the next note arrives. The Gigue’s rapid repeated arpeggios produce regular dwell phases within each figure. Speed paradoxically creates regularity. Stillness creates activity. The iterator hears the difference.

The Stylometric Portrait

Metric	Jabberwocky	Ode to Autumn	Clair de Lune	Bach Suite
Mean overhead	0.664x	0.654x	0.667x	0.657x
D* std deviation	1.68e-04	1.17e-04	1.49e-05	1.32e-04

Clustered entries	42.9%	6.7%	66.7%	56.5%
High basin %	75%	83%	—	78%
Mean depth (N)	258	352	97	71
Type-token ratio	0.545	0.693	n/a	n/a

The portrait separates cleanly along two axes.

Clustering ranks the works by their internal repetition structure: Clair de Lune (66.7%) > Bach (56.5%) > Jabberwocky (42.9%) > Keats (6.7%). Musical formal structure produces the highest clustering. Romantic poetry the lowest.

*D std deviation** ranks the works by their attractor tightness: Clair de Lune ($1.49e-05$) < Keats ($1.17e-04$) < Bach ($1.32e-04$) < Carroll ($1.68e-04$). Debussy's tonal consistency produces the tightest clustering. Carroll's varied rhyme scheme the widest spread.

The two axes are independent. Jabberwocky clusters heavily but spreads widely — repetitive but phonologically varied. Keats clusters rarely but spreads narrowly — unique but metrically consistent. The construction captures both dimensions simultaneously from the same arithmetic.

What Is Being Recovered

These are not surface similarities. The construction performs no string matching, no frequency counting, no edit distance calculation. It applies no music-theoretic rules, no phonological alphabet, no trained similarity model.

What it recovers is arithmetic equivalence under the iterator's dynamics. Two payloads share a D^* value when their bit sequences drive the same contractive attractor to the same terminal state — when they are geometrically equivalent in the iterator's phase space.

That equivalence corresponds to:

- **Exact repetition:** Carroll's repeated stanza, Debussy's recapitulation, Bach's returning motifs — same bits, same digest
- **Motivic equivalence:** Debussy's descending answer sharing D^* with the rising sequence — opposite motion, same arithmetic destination
- **Cross-movement unity:** Bach's Allemande material recurring in Menuet I — the suite's internal architecture made visible

- **Metrical regularity:** Keats' iambic lines clustering in D^* space — same rhythmic geometry
- **Rhyme families:** Carroll's and Keats' end-rhymes collapsing to equivalence classes — shared suffix bit patterns

The structure is made manifest.

The Hofstadter Resonance

Hofstadter chose Bach, Escher, and Gödel for his exploration of self-reference and strange loops. Bach's fugues are the musical paradigm of self-referential structure — the subject transforming itself through inversion, retrograde, augmentation. The Cello Suite No. 1 exhibits the same motivic economy across six movements. The iterator finds it.

The quadratic iterator is itself a strange loop: a construction designed to encode information discovers, in the process of encoding, the structural identity of what it is encoding. It was not designed to hear the recapitulation. It was not designed to feel the diminuendo. It was not designed to find Bach's cross-movement motivic unity. It was not designed to detect the rhyme.

It found these things because they were always there — in the arithmetic geometry of the bit stream, waiting for a contractive map at the right precision floor to make them visible.

The residual is not error. It is the signal. And tonight it played Clair de Lune and the Cello Suite.

Honest Scope

The musical encoding uses full note names — unambiguous across all phrases. The encoding is still reduced: microtonal inflection, dynamics, equal temperament relationships, and timing are not present. The phrase boundaries are manually defined. A production system would require automated segmentation and a richer encoding, potentially full MIDI byte sequences.

Null model validation. Clustering rates were validated against 500 random string trials matching each corpus in phrase count, length distribution, and character set. All four works exceed the null distribution at $Z > 15$ ($p < 0.001$). Random strings of equivalent structure cluster at 0–3.4%. The real works cluster at 6.7–66.7%. The structural signal is not noise.

Sarabande qualification. The Sarabande's near-zero D^* std deviation ($3.26e-08$) is

enhanced by space characters encoding rests — the spaces contribute to arithmetic convergence. Without spaces the std deviation rises to $1.93e-04$ (normal range). The structural repeat $P25==P29$ is confirmed in both encodings. The clustering is real; the near-zero std deviation is partially a spacing effect.

Key signature test. Harmonically unrelated phrases in G major show D^* distances of $1e-4$ to $3e-4$ — they do not cluster together. The metric detects motivic structure, not key signature alone.

Node-edge traversal. The digest chain was verified: all nodes independently confirmed, all edges intact ($digest[n] == seed[n+1]$ to 14dp), all entries self-certifying via fixed-point loop. The index is reproducibly traversable. Full verification code is available at unitambo.wordpress.com.

Bijectivity. Formal proof of bijectivity at fixed iterative depth remains an open question — identified as the foundational next step in the main paper. The empirical results — zero collisions across all tested payloads, exact structural detections, reproducible node-edge traversal — are consistent with bijectivity but do not constitute proof. This is stated honestly and the construction's claims do not require the formal proof to stand.

Open Questions

Can the construction detect key signature as an attractor region? Does G major cluster differently from D-flat major across multiple works?

Does the four-channel quad comparator (forward/reverse/complement/reverse complement) detect musical inversion and retrograde — the compositional techniques Bach used systematically in his fugues?

Can the J-stream KL divergence distinguish musical styles — Baroque counterpoint from Impressionist harmony — as it distinguishes Carroll from Keats?

Does the fixed-point loop convergence rate vary with musical complexity — does a Bach fugue converge differently from a Debussy nocturne?

These questions are tractable with the existing construction. No new machinery required.

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“The arithmetic heard the recapitulation. The iterator felt the diminuendo. The residual is not error — it is the signal.”